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The Sensitivity of College Football Rankings
to Several Modeling Choices

Andrew T. Karl, *Arizona State University*

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Abstract

This paper proposes a multiple-membership generalized linear mixed model for ranking college football teams using only their win/loss records. The model results in an intractable, high-dimensional integral due to the random effects structure and nonlinear link function. We use recent data sets to explore the effect of the choice of integral approximation and other modeling assumptions on the rankings. Varying the modeling assumptions sometimes leads to changes in the team rankings that could affect bowl assignments.

KEYWORDS: BCS, EM algorithm, fully exponential Laplace approximation, generalized linear mixed model, multiple membership

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1 Introduction

The highest level of collegiate football, the Football Bowl Subdivision (FBS, historically Division I-A), is undergoing a redesign of its postseason structure. Currently, the FBS is unique in that its season does not end with a tournament to decide a champion. After the regular season, the top performing teams are invited to one of several “bowl games.” Through 1997, the bowl assignments were strictly a function of conference membership (with special rules for non-conference teams), meaning the top two ranked teams often played in different bowl games. After the bowl games, the national champion was decided by the Associated Press and the Coaches polls. The championship was split when the polls disagreed. To prevent this, starting in 1998 the Bowl Championship Series (BCS) began to seed the Fiesta, Orange, Rose, and Sugar Bowls with certain conference champions and other highly ranked teams. The ranking procedures used by the BCS have changed since 1998, but have always been a weighted average of two polls and an average of several mathematical models, referred to as “computer rankings”. The computer rankings were originally allowed to use margin of victory, but in an attempt to prevent teams from running up scores to inflate their ratings, only win/loss information may now be used (ties are necessarily settled in overtime). This restriction to a binary response complicates the ranking process. Analyzing the binary game outcomes requires modeling assumptions and decisions that are not needed when modeling continuous outcomes, such as margin of victory. We explore the sensitivity of team rankings to these assumptions.

Stern et al. (2004) provide a detailed history of college football rankings as well as descriptions of the models currently employed by the BCS. An additional literature review is provided by West and Lamsal (2008). Stern et al. (2004) discuss popular controversies surrounding the BCS system through the beginning of the 2004 season, and Stern (2006) calls for a boycott of the BCS by quantitative analysts. Due in part to widespread criticism (and perhaps in larger part to potential television revenue), the current BCS structure will be revised to include a four team playoff beginning with the 2014 season. The tournament will be seeded by a selection committee, which may choose to factor mathematical rankings into their decision. A similar selection committee uses the Ratings Percentage Index (RPI) and proprietary models of Jeff Sagarin to help seed the NCAA Men’s Division I Basketball Championship (West, 2006).

There are several mathematical models available for ranking teams. One approach for continuous outcomes is to model a rating η_i for each team i , and compute the predicted win margin of team i over team j via a function

of the difference $\eta_i - \eta_j$. Such models require the margin of victory from past games. Harville (1977a) and Harville (2003) develop such a model for continuous responses, and provide methods for limiting the utility of running up the score beyond a threshold C by either truncating the win margins at C , or by scaling the margins that exceed C via a hazard function. Gill and Keating (2009) examine differences in the rankings resulting from treating the team effects as fixed or as random, as well as the modifications proposed by Harville (2003).

Despite the existence of models that minimize the advantage of running up the score, such as those proposed by Harville (2003), the BCS has decided to use only win/loss information in the computer rankings. In order to model the probability that team i defeats team j , the Thurstone-Mosteller model (Thurstone, 1927; Mosteller, 1951) calculates $\Phi(\eta_i - \eta_j)$, where Φ is the cumulative distribution function of the standard normal distribution. This is similar to the Bradley-Terry model (Bradley and Terry, 1952) used by Keener (1993). However, all of these models encounter an infinite likelihood if any teams have a perfect record, due to quasi-complete separation of the data (Allison, 2008). Mease (2003) proposes a penalized likelihood approach which circumvents the difficulty associated with the presence of undefeated or winless teams. In essence, the fixed effect model proposed by Mease (2003) becomes a random effects model with his introduction of a penalty function. Modeling the team ratings η_i with random instead of fixed effects avoids the problem of complete or quasi-complete separation. In this case, the empirical best linear unbiased predictors (EBLUPs) of the random effects are sorted to form the team rankings.

The penalized likelihood used by Mease (2003) requires a choice of a penalty function. Annis and Craig (2005) express concern that this subjective choice may influence the rankings produced by the model. The model proposed by Mease (2003) may be seen as a special case of the generalized linear mixed model (GLMM) we propose: it arises as one particular approximation of the marginal likelihood of our GLMM. Furthermore, Mease (2003) uses a probit link, and mentions the possibility of using a logit link as an alternative, noting that the choice between the two “did not affect the resulting rankings substantially for the football seasons considered.”

The choice of link function is often minimized in discussions of generalized linear mixed models, and the choice of integral approximation depends on computational feasibility as determined by the structure of the random effects. It is important to note that, even if the true parameter values were known, the EBLUPs in a GLMM depend on the chosen integral approximation. Finally, it is well known that maximum likelihood (ML) estimates for variance compo-

nents are subject to a downward bias, and that restricted maximum likelihood (REML) estimation procedures correct for this bias (Harville, 1977b). In this paper we explore how these modeling choices affect the team rankings.

We present a GLMM for the ranking of college football teams using only win/loss information. Our model structure is the same as the one proposed by Mease (2003), except we account for the teams via random instead of fixed effects. Furthermore, we consider treating the FBS and FCS (Football Championship Subdivision, formerly Division I-AA) divisions as different populations, whereas Mease (2003) consolidates all FCS teams into a single effect that is regarded as member of the FBS population. We show that our GLMM is a generalization of the penalized likelihood proposed by Mease (2003), and explore the sensitivity of the rankings from our model to the choices of

1. link function
2. integral approximation used for the marginal likelihood of our GLMM
3. distribution of the random team effects
4. consolidating FCS teams into a single “team” vs. treating FCS as a separate population
5. ML vs. REML

The results from our analysis show that the changes in team ratings resulting from varying these assumptions are small relative to the standard errors associated with each team’s rating. However, these changes result in a reordering of the team rankings that may lead to practically significant differences in the bowl selection process. There is limited information in the binary win/loss outcome of each game, and it is not surprising that changing properties of the model may lead to teams with similar records swapping ranks. For some of the assumptions listed above, especially points 1 and 3, there may not be a clear set of best choices. However, these choices may affect the ordering of the team rankings.

We present the model in Section 2 and discuss the parameter estimation in Section 3. In Section 4, we compare the rankings resulting from our model under different assumptions for the seasons 2008-2011. In each year, we use data through the end of the conference championships, excluding the outcomes of the bowl games. Thus our examples use the same data that were used to compile the final BCS rankings in each year, which were used as a basis for the bowl invitations.

2 The Model

Mease (2003) models team ratings with fixed effects $\boldsymbol{\theta} = (\theta_1, \dots, \theta_{p+1})$ via the likelihood

$$l(\boldsymbol{\theta}) = \prod_{(i,j) \in S} [\Phi(\theta_i - \theta_j)]^{n_{ij}} \quad (1)$$

$$\times \prod_{i=1}^p \Phi(\theta_i) \Phi(-\theta_i) \quad (2)$$

$$\times \Phi(\theta_{p+1}) \Phi(-\theta_{p+1}) \times \prod_{(i,j) \in S^*} [\Phi(\theta_i - \theta_j)]^{n_{ij}} \quad (3)$$

where S is the set of all ordered pairs (i, j) for which team i defeated team j , and both teams belong to the FBS. p is the number of FBS teams, and θ_{p+1} is a single effect that is used to represent every FCS team. S^* is defined in the same way as S , except one of the teams in each pair is from the FBS and the other is from the FCS. Mease (2003) refers to Equations (1)-(3) as Parts 1, 2, and 3, respectively. Part 1 models the probability of the outcome of each game using the team ratings, and implicitly considers “strength of schedule.” The second part is a penalty function that allows the model to be estimated in the presence of undefeated or winless teams: using Part 1 alone leads to an infinite likelihood in these cases. The third and final part models games between FBS and FCS teams using a single team effect to represent all FCS teams.

We propose modeling the team ratings with random instead of fixed effects, and show that the model proposed by Mease (2003) is actually a special case of our random effects model. Treating the teams as random effects requires the specification of a distribution for those effects, as well as the choice of an integral approximation for the resulting generalized linear mixed model. In addition, we propose another method of modeling games between FBS and FCS teams. Besides modeling all FCS teams as a single effect and ignoring FCS games that were not played against an FBS opponent, we model FCS teams as a separate population. Using this approach, we include all of the games played between two FCS teams, but ignore FCS games played against lower-divisional opponents. This introduces two more modeling assumptions. 1) Instead of ignoring FCS games against lower divisional opponents, those opponents could be treated as a single effect, using the same approach as Mease (2003). Although we do not take this approach, this could protect against the possibility of a successful FCS team being overrated due to an ignored loss against a Division II team. 2) When modeling separate FBS and

FCS populations, we may either assume that the populations share a common variance, or we may model a different variance for each population. In Section 4, we model both pooled and separate variances and compare the resulting differences.

2.1 Separate FCS Population with Pooled Variance

We first present our model including separate FBS and FCS populations, assuming that the FBS and FCS distributions share a common variance. In Section 2.2 we describe the changes necessary for modeling different FBS and FCS effect variances, and in Section 2.3 we treat FCS opponents of FBS teams as a single team in the FBS population

Our model considers outcomes of FBS and FCS games in a given season. Let r_i be a binary indicator for the outcome of the i -th game for $i = 1, \dots, n$, taking the value 1 with a home team win and 0 with a visiting team win. For neutral site games, designate a home team arbitrarily.

We model the rating of the j -th team for $j = 1, \dots, p + q$ with a random effect η_j assuming $\eta_j \sim N(0, \sigma_t^2)$. p and q represent the number of included FBS and FCS teams, respectively. We assume that the distributions of FBS and FCS ratings share a common variance σ_t^2 , but that the distributions have different means. We account for the difference in means between the two divisions by including the fixed effect β in the model. The coefficient X_i for β takes the value 1 if the i -th game involves an FCS team visiting an FBS team, and 0 otherwise (FBS teams do not travel to play FCS teams). We will refer to β as the “FCS effect.”

Using the threshold model of McCulloch (1994), we assume that the game outcomes are determined by a latent continuous variable $y_i = X_i\beta + \mathbf{Z}_i\boldsymbol{\eta} + \epsilon_i$, which may be interpreted as the margin of victory for the home team, but only the binary outcome $r_i = I_{\{y_i > 0\}}$ is observed. We thus model the probability π_i of a home team win in the i -th game, as.

$$\pi_i = P(X_i\beta + \mathbf{Z}_i\boldsymbol{\eta} + \epsilon_i > 0)$$

where β is the FCS effect, $\boldsymbol{\eta} = (\eta_1, \dots, \eta_{p+q}) \sim N(0, \sigma_t^2 \mathbf{I})$ contains the random effects representing the team ratings, $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_n) \sim N(0, \mathbf{I})$, and $\text{cov}(\boldsymbol{\eta}, \boldsymbol{\epsilon}) = 0$. The assumed distribution of ϵ determines the link function of the resulting GLMM. Assuming that $\boldsymbol{\epsilon} \sim N(0, \mathbf{I})$ leads to a probit link,

$$\begin{aligned} r_i | \boldsymbol{\eta} &\sim \text{Bin}(1, \pi_i) \\ \Phi^{-1}(\pi_i) &= X_i\beta + \mathbf{Z}_i\boldsymbol{\eta} \end{aligned}$$

By contrast, assuming that the ϵ_i are independent and follow a logistic distribution leads to a logit link.

$$r_i | \boldsymbol{\eta} \sim \text{Bin}(1, \pi_i)$$

$$\text{logit}(\pi_i) = X_i \beta + \mathbf{Z}_i \boldsymbol{\eta}$$

where $\text{logit}(\pi) = \log(\pi/(1 - \pi))$.

The design matrix $\mathbf{Z}$ for the random effects contains rows $\mathbf{Z}_i$ that indicate which teams competed in game i . If team k visits team l in game i , then $\mathbf{Z}_i$ is a vector of zeros with a 1 in position l and a -1 in position k . This is a multi-membership design (Browne et al., 2001) since each game belongs to multiple levels of the same random effect. As a result, $\mathbf{Z}$ does not have a patterned structure and may not be factored as it could be with nested designs. Mease (2003) uses the same design matrix, albeit using fixed effects for the teams. The multi-membership $\mathbf{Z}$ matrix is implied by the structure of the product over the set S in Equation (1).

We choose not to include a home field effect. We are concerned about the potential impact of nonrandom scheduling between teams on the estimate of home field advantage. Only three of the 120 FBS college football teams (Notre Dame, UCLA, and Southern California) have refrained from scheduling matches against FCS teams to lighten their schedules. These games are accounted for with the FCS effect, but there is yet another concern about the estimation of a home field effect. The more competitive programs tend to play more home games than other FBS programs. These additional home games are often against low-level FBS teams who do not possess the bargaining leverage to request a home-and-home series. As a result, the “home field advantage” may appear to be significant simply because of the tendency of larger schools to schedule a number of easier home games. Others have advocated including a home field effect, including Harville (1977a), Harville (2003), Mease (2003), and Wang et al. (2011).

Other fixed effects may easily be included in the model, but the factors should not contain any levels that act as perfect predictors for the outcomes. For example, the inclusion of the FCS fixed effect will lead to an infinite likelihood due to quasi-complete separation (Allison, 2008) in the event that every FBS vs. FCS game results in an FBS win. In 2008, FCS teams won only 2 of 86 games against FBS teams. In the event that FBS teams were to win all such games in a year, the FCS effect and teams may be removed from the model. Historically, some of the BCS computer rankings always ignored these inter-division matches, until fifth-ranked Michigan lost to FCS Appalachian State after paying the team \$400,000 for a one-off home game in 2007.

The likelihood for our model with a probit link is

$$L(\beta, \sigma_t^2) = \int \cdots \int \prod_{i=1}^n [\Phi((-1)^{1-r_i} [X_i\beta + \mathbf{Z}_i\boldsymbol{\eta}])] f(\boldsymbol{\eta}) d\boldsymbol{\eta} \quad (4)$$

where $f(\boldsymbol{\eta})$ is the density of $\boldsymbol{\eta}$. We assume $\boldsymbol{\eta} \sim N_{p+q}(0, \sigma_t^2 \mathbf{I})$. Using a logit link,

$$L(\beta, \sigma_t^2) = \int \cdots \int \prod_{i=1}^n \left(\frac{e^{X_i\beta + \mathbf{Z}_i\boldsymbol{\eta}}}{1 + e^{X_i\beta + \mathbf{Z}_i\boldsymbol{\eta}}} \right)^{r_i} \left(\frac{1}{1 + e^{X_i\beta + \mathbf{Z}_i\boldsymbol{\eta}}} \right)^{1-r_i} f(\boldsymbol{\eta}) d\boldsymbol{\eta}. \quad (5)$$

The model likelihood functions in Equations (4) and (5) contain intractable integrals because the random effects enter the model through a nonlinear link function. Furthermore, the $(p+q)$ -dimensional integral in each equation may not be factored as a product of one-dimensional integrals. Such a factorization occurs in longitudinal models involving nested random effects. However, the multi-membership random effects structure of our model results in a likelihood that may not be factored.

2.2 Separate FCS Population with Unique Variance

The model assuming separate FBS and FCS populations uses the same setup as described in Section 2.1, except that the populations are allowed to have different variances. The likelihood function for this model with a probit link is

$$L(\beta, \sigma_1^2, \sigma_2^2) = \int \cdots \int \prod_{i=1}^n [\Phi((-1)^{1-r_i} [X_i\beta + \mathbf{Z}_i\boldsymbol{\eta}])] f(\boldsymbol{\eta}) d\boldsymbol{\eta}$$

where $f(\boldsymbol{\eta})$ is the density of $\boldsymbol{\eta} = (\boldsymbol{\eta}_1, \boldsymbol{\eta}_2)$, with $\boldsymbol{\eta}_1$ and $\boldsymbol{\eta}_2$ containing the FBS and FCS team effects, respectively. We assume $\boldsymbol{\eta}_1 \sim N_p(0, \sigma_1^2 \mathbf{I})$, $\boldsymbol{\eta}_2 \sim N_q(0, \sigma_2^2 \mathbf{I})$, and that $\text{cov}(\boldsymbol{\eta}_1, \boldsymbol{\eta}_2) = 0$. Using a logit link,

$$L(\beta, \sigma_1^2, \sigma_2^2) = \int \cdots \int \prod_{i=1}^n \left(\frac{e^{X_i\beta + \mathbf{Z}_i\boldsymbol{\eta}}}{1 + e^{X_i\beta + \mathbf{Z}_i\boldsymbol{\eta}}} \right)^{r_i} \left(\frac{1}{1 + e^{X_i\beta + \mathbf{Z}_i\boldsymbol{\eta}}} \right)^{1-r_i} f(\boldsymbol{\eta}) d\boldsymbol{\eta}.$$

2.3 Single Population

The model consolidating FCS teams into a single “team” in the FBS population also uses a similar setup as described in Section 2.1. However, this model does not require the FCS fixed effect, and discards information about games

played between pairs of FCS teams. The likelihood function using a probit link is

$$L(\sigma_t^2) = \int \cdots \int \prod_{i=1}^n [\Phi((-1)^{1-r_i} \mathbf{Z}_i \boldsymbol{\eta})] f(\boldsymbol{\eta}) d\boldsymbol{\eta} \quad (6)$$

where $f(\boldsymbol{\eta})$ is the density of $\boldsymbol{\eta}$. We assume $\boldsymbol{\eta} \sim N_{p+1}(0, \sigma_t^2 \mathbf{I})$, where η_{p+1} is consolidated FCS team-effect. Using a logit link,

$$L(\sigma_t^2) = \int \cdots \int \prod_{i=1}^n \left(\frac{e^{\mathbf{Z}_i \boldsymbol{\eta}}}{1 + e^{\mathbf{Z}_i \boldsymbol{\eta}}} \right)^{r_i} \left(\frac{1}{1 + e^{\mathbf{Z}_i \boldsymbol{\eta}}} \right)^{1-r_i} f(\boldsymbol{\eta}) d\boldsymbol{\eta}.$$

The model proposed by Mease (2003) results from applying a particular integral approximation to Model (6). Following an illustration by Demidenko (2004), the penalized likelihood used by Mease (2003) may be derived from our model (6) via the Laplace approximation (Evans and Swartz, 1995). Letting

$$h(\boldsymbol{\eta}) = \log \left[\prod_{i=1}^n [\Phi((-1)^{1-r_i} \mathbf{Z}_i \boldsymbol{\eta})] f(\boldsymbol{\eta}) \right],$$

the Laplace approximation yields

$$L(\sigma_t^2) \approx (2\pi)^{n/2} e^{h(\boldsymbol{\eta}^*)} \left| -\frac{\partial^2 h}{\partial \boldsymbol{\eta} \partial \boldsymbol{\eta}'} \right|_{\boldsymbol{\eta}=\boldsymbol{\eta}^*}^{-1/2} \quad (7)$$

where $\boldsymbol{\eta}^*$ is the mode of $h(\boldsymbol{\eta})$. Further assuming that the determinant in Equation (7) varies slowly with $\boldsymbol{\eta}$ yields the quasi-likelihood (Breslow and Clayton, 1993)

$$L(\boldsymbol{\eta}) \approx \prod_{i=1}^n [\Phi((-1)^{1-r_i} \mathbf{Z}_i \boldsymbol{\eta})] f(\boldsymbol{\eta}) \quad (8)$$

If the random effects $\boldsymbol{\eta}$ are assumed to be distributed so that

$$f(\boldsymbol{\eta}) \propto \prod_{j=1}^{p+1} \Phi(\eta_j) \Phi(-\eta_j), \quad (9)$$

then Equation (8) yields the likelihood presented by Mease (2003) in Equations (1-3). Thus the model of Mease (2003) may be viewed as the PQL approximation to our probit model (6), where the random effects are assumed to have the density specified in Equation (9) rather than the normal density that we specified.

3 Parameter Estimation

Section 2.3 demonstrates that the model likelihood of Mease (2003) is the PQL approximation to our model (6) under a certain non-normal distribution of the random effects, $\boldsymbol{\eta}$. PQL is based on the Laplace approximation, but makes one further approximation, and produces biased parameter estimates (Breslow and Lin, 1995). By contrast, the Laplace approximation produces consistent estimators since the dimension of our integrals is equal to the number of teams and does not increase with the sample size (Shun and McCullagh, 1995). In Section 3.1 we demonstrate how our model may be fit in SAS using a PQL approximation, and in Section 3.2 we explain how we use an EM algorithm with both a first-order and a fully exponential Laplace approximation to obtain the rankings. Code for fitting the model proposed by Mease (2003) is available from Mease (2012).

3.1 Fitting the Model in SAS

Specifying the multi-membership random effects structure in SAS is possible in PROC GLIMMIX via the MULTIMEMBER option of the EFFECT statement. However, GLIMMIX does not take into account the fact that $\mathbf{Z}$ is sparse. This is not an issue for the football data sets used here, but becomes problematic in other settings with larger data sets. The EM algorithm we use in Section 3.2 provides computational advantages in the estimation of multi-membership models (Karl et al., 2012).

The default estimation method of PROC GLIMMIX is a doubly-iterative pseudo-likelihood method by which the link function is linearized and a linear mixed model is fit at each iteration. This method is equivalent to PQL when the scale parameter is set to 1, which is the default setting for the Bernoulli distribution in PROC GLIMMIX (Wolfinger and O’Connell, 1993; Littell et al., 2006). Example SAS code appears in Appendix B.

3.2 Fitting the Model with an EM Algorithm

The EM algorithm (Dempster et al., 1977; McLachlan and Krishnan, 2008) is often used for maximum likelihood estimation in the presence of missing data. It may be used for the estimation of mixed models by treating the random effects as missing observations (Laird and Ware, 1982), although an integral approximation is necessary when the random effects enter the model through a nonlinear link function, such as is the case with our model. Note that the high dimension of the integral renders quadrature methods infeasible.

The use of a fully exponential Laplace approximation (Tierney et al., 1989) with an EM algorithm for the estimation of generalized linear mixed models was first proposed by Steele (1996). Rizopoulos et al. (2009) use this method to estimate the parameters of a joint model for a continuous longitudinal process and a time-to-dropout measurement. Karl (2012a) applies this approach to a multi-membership joint model. We will give a brief overview of the EM estimation procedure, and refer to Karl (2012a) for further details, as well as to Rizopoulos et al. (2009) for similar calculations made in the setting of nested random effects.

As an alternative to the PQL approximation, we fit the model using an EM algorithm with custom-written code in R (R Development Core Team, 2012) with both a first order Laplace (LA) and a fully exponential Laplace (FE) approximation in the E-step. The Laplace approximations are more accurate than PQL, but are more computationally demanding, requiring the calculation of higher-order derivatives than PQL. The first order Laplace approximation requires the first two derivatives of the integrand. Calculation of the fully exponential Laplace approximation for the conditional mean of the random effects requires the third derivative, and calculation of the conditional variance requires the fourth derivative. The approximation is complicated by the multi-membership random effects structure.

We outline the EM procedure for only one of the models, and note that the calculations are similar for the other models. To estimate the parameters $\Psi = (\beta, \sigma_t^2)$ of the model in Equation (4), we use the equations derived by Karl (2012a), ignoring the longitudinal process in the joint model he analyzed. Given initial values for the parameters and the random effects, the EM algorithm alternates between an expectation (E) step and a maximization (M) step. At iteration $(k + 1)$, the E step calculates the conditional expectation of the log-likelihood $\log f(\mathbf{r}, \boldsymbol{\eta})$,

$$Q(\Psi; \Psi^{(k)}) = \int \cdots \int \{\log f(\mathbf{r}|\boldsymbol{\eta}; \Psi) + \log f(\boldsymbol{\eta}; \Psi)\} f(\boldsymbol{\eta}|\mathbf{r}; \Psi^{(k)}) d\boldsymbol{\eta},$$

given the vector of game outcomes, $\mathbf{r}$, and parameter estimates $\Psi^{(k)}$ obtained in the k -th iteration. For this calculation, it is sufficient to find the conditional mean $\tilde{\boldsymbol{\eta}} = E[\boldsymbol{\eta}|\mathbf{r}; \Psi^{(k)}]$ and the conditional variance $\tilde{\mathbf{v}} = \text{var}[\boldsymbol{\eta}|\mathbf{r}; \Psi^{(k)}]$ of the random effects. The M-step then maximizes $Q(\Psi; \Psi^{(k)})$ with respect to Ψ , resulting in the updated parameter vector $\Psi^{(k+1)}$.

The expressions for $\tilde{\boldsymbol{\eta}}$ and $\tilde{\mathbf{v}}$ involve intractable integrals, necessitating the use of the Laplace approximations. The gradient and inverse Hessian of the joint distribution $f(\mathbf{r}, \boldsymbol{\eta})$ (with respect to $\boldsymbol{\eta}$) furnish the first order Laplace approximations to $\tilde{\boldsymbol{\eta}}$ and $\tilde{\mathbf{v}}$, respectively, and the fully exponential Laplace

approximations require computationally expensive corrections to these values. Upon convergence, $\tilde{\boldsymbol{\eta}}$ serves as the vector of team ratings, demonstrating that the choice of integral approximation affects the ratings even if the model parameters $\boldsymbol{\Psi}$ are known.

The M-step update for $\hat{\beta}$ from the probit model may be obtained by setting the score function

$$S(\hat{\beta}) = \sum_{i=1}^n X_i \int \cdots \int (-1)^{1-r_i} \frac{\phi \left[(-1)^{1-r_i} (X_i \hat{\beta} + \mathbf{Z}_i \boldsymbol{\eta}) \right]}{\Phi \left[(-1)^{1-r_i} (X_i \hat{\beta} + \mathbf{Z}_i \boldsymbol{\eta}) \right]} f(\boldsymbol{\eta} | \mathbf{r}) d\boldsymbol{\eta} \quad (10)$$

equal to 0, where ϕ is the density function of the standard normal distribution. The equation may be solved via Newton-Raphson, using a central difference approximation to $S(\hat{\beta})$ in order to obtain the necessary Hessian. Fortunately, there is a closed form M-step update for σ_t^2 , namely

$$\hat{\sigma}_t^2 = \frac{1}{p+q} \text{trace}(\tilde{\mathbf{v}} + \tilde{\boldsymbol{\eta}} \tilde{\boldsymbol{\eta}}').$$

4 Application

We obtain the game outcomes for the 2008-2011 seasons from the NCAA website (NCAA, 2012). The data require additional processing, since outcomes are recorded by teams, resulting in duplicate observations for games between two teams within the same division. Some of the neutral site games were duplicated while others were not. We combine the FBS and FCS files, remove all games labeled “away”, remove games between FCS and lower division teams, purge redundant neutral site games, add an indicator for games played between FBS and FCS schools, and remove the records of games played after the production of the final BCS rankings in each year. The processed data are available from Karl (2012b). See Table 12 in Appendix B for the first observations of our 2008 data set.

The team ratings and rankings for the 2008-2011 seasons appear in tables and scatter plots Appendix A. The scatter plots are printed with reference lines with slope 1 and intercept 0. The rankings use the game outcomes through the end of the conference championships in each year: the bowl outcomes are not included. This allows us to compare the rankings from our model to the final BCS rankings used to choose the BCS teams. The BCS rankings are included as a reference, and not as a standard that we expect our model to match. Two-thirds of the weight of the BCS ranking is given to polls, and the voters are allowed to consider more than each team’s win/loss record.

Under the current BCS configuration, the top 16 teams of the BCS rankings are relevant due to the rules involving eligibility for selection in the non-championship BCS bowls. We do not list all of the rules the selection process, but instead point out a few of the highlights. The top two teams in the BCS rankings are selected to play in the national championship game. The remaining 8 BCS slots are filled with the winners of certain conferences (e.g. Big East), regardless of their ranking. For example, after the 2010 season, #7 ranked Oklahoma was paired in the Fiesta Bowl with an (8-4) unranked Connecticut team.

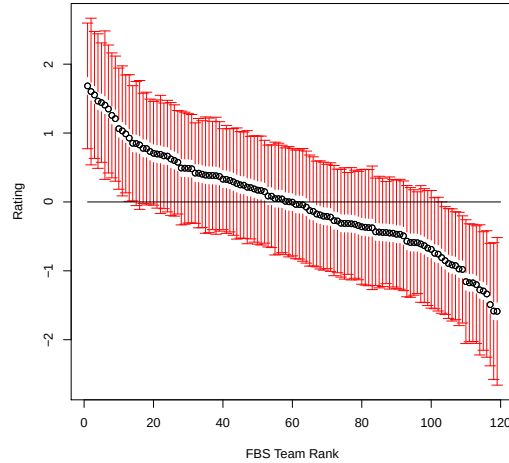
There are special rules for non-BCS teams, including a rule that the highest-ranked winner of a non-BCS conference will receive a berth if it is either ranked in the top 12 or in the top 16 and higher than at least one BCS conference winner. Under certain conditions, any team finishing in the top four is guaranteed a berth. The complete list of rules is available from the Bowl Championship Series (2011). In short, permutations in the rankings of teams outside of the top two can affect the selection of teams for participation in the BCS. The value of these berths is substantial. Each conference, and thus each school, receives an extra payout for each additional team that is awarded a BCS berth. The head coaches of these teams benefit as well, due to their contracts: Les Miles (LSU) received a \$200,000 bonus for reaching a BCS game in 2012 and would have received an extra \$100,000 for winning the national title game. In addition, a national title win for Miles would have activated a clause in his contract giving him a \$5.7 million raise over the remaining 6 years of his contract. Instead, Nick Saban (Alabama) received a \$400,000 bonus for defeating LSU (Eichelberger, 2012).

In the following subsections, we describe the changes in rankings that result from varying several modeling assumptions. For convenience, we will introduce a three-part notation to indicate the method of integral approximation (PQL, LA for Laplace, FE for fully exponential Laplace), the link function (P for probit, L for logit), and the way in which FCS teams are handled (0 for consolidating them into a single effect as in Section 2.3, 1 for modeling separate FBS and FCS populations with a pooled variance as in Section 2.1, and 2 for modeling separate populations with separate variances as in Section 2.2). For example, PQL.P.0 denotes the PQL approximation to the probit model that consolidates FCS teams into a single “team.”

4.1 The Link Function

To explore sensitivity to the choice of link function, we compare the rankings from FE.P.0 and FE.L.0, which appear in Figures 5, 6, 9, 10, 13, 14, 17, and 18

Figure 1: FE.P.0 Ratings with Standard Errors for 2008 Season.



in Appendix A. The reference lines in the scatter plots for the continuous ratings from these two models are not meaningful since the kurtosis of the logistic distribution is greater than that of the normal distribution. We would expect the rankings using the fully exponential Laplace approximation to be the most sensitive to the choice of link function, since the approximation makes use of four derivatives of the complete-data likelihood, and thus the link function.

In 2008 these two models agree on ranks 1-16, though 17-20 are scrambled by one or two positions. In 2009, 13-14 and 17-18 each swap positions. In 2010, 7 and 9 swap, and 17-20 each differ by one position. However, in 2011, FE.P.0 ranks LSU, Alabama, and Oklahoma St. as the top three teams, respectively, while FE.L.0 picks LSU, Oklahoma St., Alabama. Figure 18 shows that Alabama and Oklahoma St. have nearly identical ratings in each of the models. The change in link function leads to slight changes in the team ratings, prompting shifts in the rankings. These changes are small relative to the standard errors of the ratings, as can be seen from the caterpillar plot in Figure 1, which displays the 2008 FBS team ratings from FE.P.0 along with their associated 95% prediction intervals.

In the context of teacher evaluation, Draper (1995) urges caution when using the EBLUP rankings because individual ratings may have large standard errors. Given the high stakes involved in the BCS rankings and the way the models are used by the BCS, small changes in the ratings may be practically significant, even if they are not statistically significant. The BCS uses the discretized rankings from each model, not the continuous ratings. The plots

in Appendix A show how uneven the spacing between team ratings can be. Teams with similar ratings are more likely to swap positions with changes in the modeling assumptions.

4.2 The Integral Approximation

The tables in Appendix A contain the team rankings and ratings, as well as the parameter estimates for σ_t^2 from models PQL.P.0, LA.P.0, and FE.P.0. The downward bias in the PQL estimates of σ_t^2 described by Breslow and Lin (1995) is clearly present. The fully exponential Laplace approximation is more accurate than the first order Laplace, which is in turn better than PQL. It is interesting to see in the tables that the changes in a team's ranking from PQL to LA to FE are monotonic. That is, for 2008-2011, if a team is ranked X by PQL.P.0 and Y by FE.P.0, its ranking under LA.P.1 is somewhere between X and Y, inclusive.

Two of the more interesting changes in ranking seen across the models for these data are in 2009 and 2011, where the change in integral approximation alters which team is voted second, and thus one of the two teams to play for the national championship. In 2009, FE.P.0 lists Alabama and Cincinnati, while PQL.P.0 and LA.P.0 select Alabama and Texas. In 2011, PQL.P.0 and LA.P.0 rank Oklahoma St. 2, while FE.P.0 ranks Alabama 2. Of course, these teams only moved one position between second and third in these rankings, but the exchange is remarkable in the sense that it arises solely from using a fully exponential (asymptotically analogous to a second-order) instead of a first-order Laplace approximation. At least in these years, our answer to the question, "Who do you think should play in the BCS championship game?" depends on the response to the question, "To what order would you prefer to extend your Laplace approximation?"

Changing from PQL.P.0 to FE.P.0 in 2008 moves Texas Tech from 6 to 4, Boise St. from 5 to 6, Oklahoma St from 16 to 14, plus seven other changes of one position. Besides the Alabama/Cincinnati swap in 2009, four other teams change rank: Oregon St. moves from 23 to 20, Arizona from 21 to 19, and LSU switched 12 for 11 with Virginia Tech. In 2010, Stanford improves from 5 to 4, displacing Oklahoma. Wisconsin moves from 8 to 6, Ohio St. from 6 to 7, Arkansas from 10 to 8, Michigan St. from 7 to 9, and Boise St. from 9 to 10. Finally, in addition to the Alabama/Oklahoma St. switch, the change to the fully exponential model in 2011 lifts Arkansas from 8 to 6. Baylor moves from 16 to 14, Oklahoma from 14 to 12, Michigan from 13 to 15, Georgia from 17 to 16, and Wisconsin from 15 to 17. There were a couple of other teams that changed a single rank.

It is difficult to tell whether there is a pattern in set of teams which are affected by the choice of FE.P.0 over PQL.P.0 in each year. SEC teams tend to benefit from the change. It is possible that the increased random team-effect variance estimated by the fully exponential model implicitly places a greater emphasis on strength of schedule, since the larger the variance of the team effect, the greater the estimated disparity between FBS teams. We consider this point further in Section 4.3.

4.3 Distribution(s) of the Random Team-Effects

Annis and Craig (2005) express concern about potential sensitivity of team rankings to the penalty function chosen by Mease (2003). In Section 2.3, we demonstrated that the penalty function (2) corresponds to the distribution of the random team-effects. Figure 2 compares the normal distribution to the distribution implied by the penalty function of Mease (2003). Figure 3 compares the rankings from PQL.P.0 and the model by Mease (2003). We summarize the changes to the top five teams when moving from PQL.P.0 to Mease (2003). In 2008, Texas Tech moves from 5 to 4 and Florida moves from 4 to 5. In 2009, Texas moves from 2 to 3 and Cincinnati moves from 3 to 2. In 2010, Stanford moves from 5 to 4 and Oklahoma moves from 4 to 5. There are no changes in the top 5 in 2011, but there are several changes farther down in the top 20.

The distribution (2) used by Mease (2003) is fixed and does not depend upon estimated parameters. It is very similar to a $N(\mathbf{0}, 0.815 * \mathbf{I})$ distribution. In this sense, Mease's model may be approximated using PQL.P.0 by restricting $\sigma_t^2 = .815$. To address the point made by Annis and Craig (2005), we fit PQL.P.0 using different fixed values of σ_t^2 . At one extreme, consider the case $\sigma_t^2 = 0$. This implies that all of the teams are of equal strength, that the chance of a team winning any given game is 50%, and that the teams should be ranked by their number of wins minus their number of losses. To explore this possibility, we restricted $\sigma_t^2 = 0.0001$ in our R program (which requires a positive value of σ_t^2) and found roughly what we expected in the left column of Table 1. Notice how Arkansas St. made it up to 11 under this scenario. For the other extreme, we restricted $\sigma_t^2 = 100$. Notice how 12 of the top 15 teams are from either the SEC or Big XII, including 7-5 Auburn and Texas, and 6-6 Texas A&M. Note that Arkansas' two losses were to LSU and Alabama, and that Alabama's only loss was to LSU.

Our model provides an advantage over that of Mease (2003) by estimating σ_t^2 rather than fixing it at an arbitrary value. This discussion also provides at least a partial explanation for the changes in ranking due to changes of integral

approximation. The downward bias in the PQL estimates of σ_t^2 (Breslow and Lin, 1995) has the same effect as modifying the assumed distribution of the random effects. Across the years, the PQL.P.0 estimates of σ_t^2 tend to be around 0.55, while the FE.P.0 estimates tend to be closer to 0.8. This seems to explain why Mease’s model tends to agree more closely with FE.P.0 instead of PQL.P.0, despite relying on a PQL approximation.

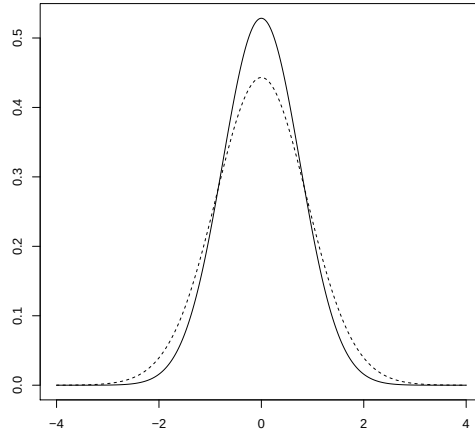
Table 1: Rankings under extreme values of σ_t^2 .

$\sigma_t^2 = 0.0001$				$\sigma_t^2 = 100$			
Rank	Team	W	L	Rank	Team	W	L
1	LSU	13	0	1	LSU	13	0
2	Alabama	11	1	2	Alabama	11	1
3	Boise St.	11	1	3	Arkansas	10	2
4	Houston	12	1	4	Oklahoma St.	11	1
5	Oklahoma St.	11	1	5	Kansas St.	10	2
6	Stanford	11	1	6	South Carolina	10	2
7	Oregon	11	2	7	Baylor	9	3
8	Southern Miss.	11	2	8	Oklahoma	9	3
9	Virginia Tech	11	2	9	Stanford	11	1
10	Arkansas	10	2	10	Oregon	11	2
11	Arkansas St.	10	2	11	Georgia	10	3
12	Clemson	10	3	12	Boise St.	11	1
13	Georgia	10	3	13	Auburn	7	5
14	Kansas St.	10	2	14	Texas	7	5
15	Michigan	10	2	15	Texas A&M	6	6

4.4 Modeling FCS Teams

We consider three different approaches to handling FBS games against FCS teams, using a fully exponential approximation with a probit link. FE.P.0 uses the same approach as Mease (2003), discarding all FCS games that did not involve an FBS opponent and consolidating all FCS teams into a single “team” in the population of FBS teams. FE.P.1 models a separate FCS population, using a pooled estimate of the FBS and FCS population variances. Finally, FE.P.2 models separate populations, estimating a different variance for each population. The results appear in the tables in Appendix A, and are also compared to Mease’s model. The reference lines with slope 1 and intercept 0

Figure 2: The random effects distribution (dashed) implicitly assumed by Mease (2003) and the $N(0, 0.57)$ distribution from PQL.P.0 in 2009 (solid).



in the scatter-plots in Appendix A illustrate the difference in estimated team-effect variances between the models. For example, the plot in position [1,2] in the matrix of plots in Figure 8 shows the ratings falling below the reference line. This indicates that, in 2008, the variance of the FBS team-effects from FE.P.2 is less than that of FE.P.1.

Table 2 shows the estimated pooled variance (σ_t^2) from FE.P.1 and the FBS variance (σ_1^2) and FCS variance (σ_2^2) from FE.P.2. Figure 4 plots the distributions of the team ratings from FE.P.1 2011. This plot illustrates the difference in means for the two populations, as well as the overlap between the distributions. This difference, obtained via the estimate $\hat{\beta} = 2.03$, indicates that the probability of a randomly selected FBS team defeating a randomly selected FCS team in 2011 is $\Phi(2.03) \approx 0.979$. FE.P.0 ranks Alabama 2 in 2011, while FE.P.1 and FE.P.2 pick Oklahoma St. In 2008, FE.P.2 ranks Florida 4, while FE.P.1 and FE.P.0 rank Texas Tech 4. There are other changes lower in the rankings as well.

FE.P.2 provides greater flexibility than FE.P.1 by estimating separate variance components. Likewise, FE.P.2 makes use of more information than FE.P.0, by considering the outcome of games between pairs FCS teams as well as modeling the FBS games against specific FCS teams rather than a generic FCS “team.” We prefer FE.P.2; however, the approach used by FE.P.0 is reasonable as well. As we have seen in previous sections, the choice between two reasonable assumptions may lead to different rankings.

Figure 3: Comparison of rankings resulting from normally distributed random effects (PQL.P.0) to those obtained under the distribution assumed by Mease (2003)

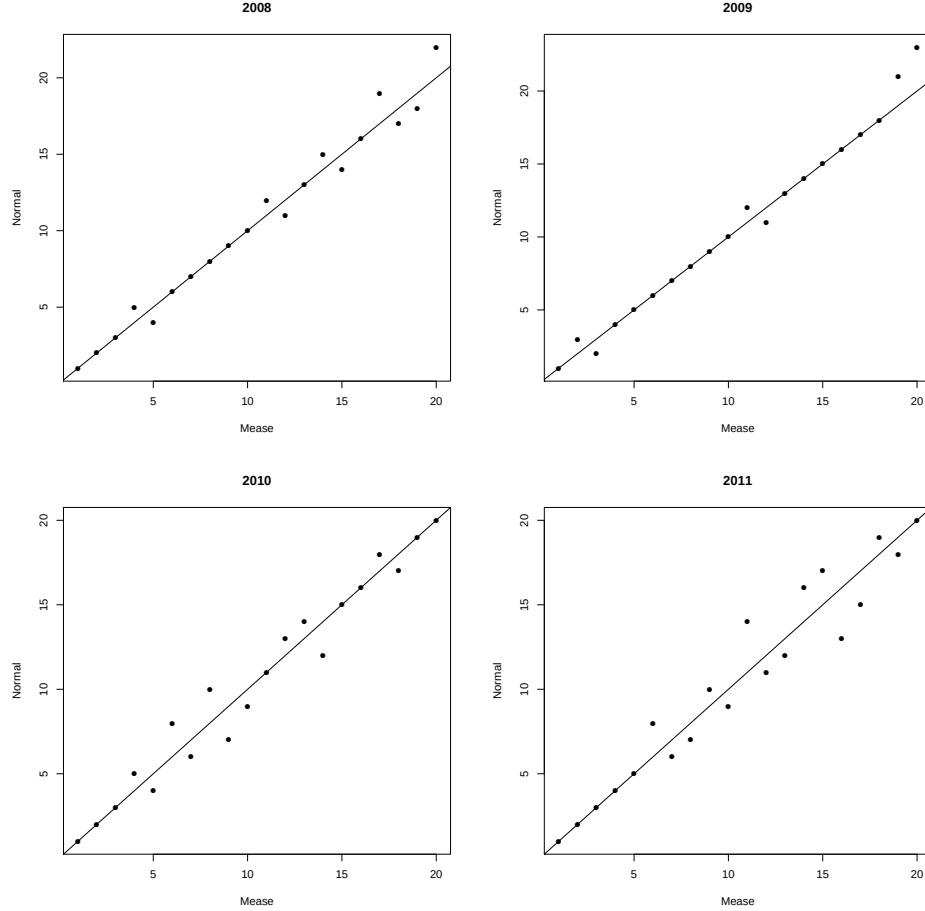
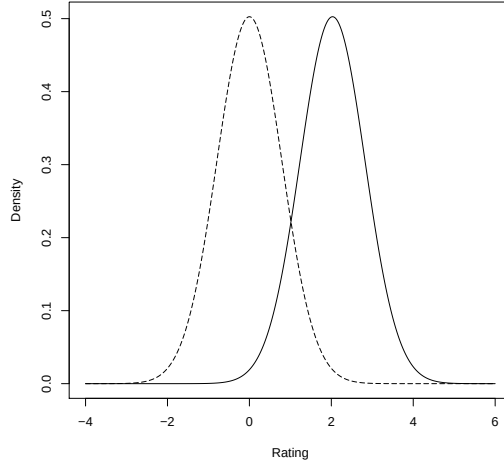


Table 2: Estimates for σ_t^2 from FE.P.1, and σ_1^2 and σ_2^2 from FE.P.2

Year	σ_t^2	σ_1^2	σ_2^2
2008	0.75	0.65	0.87
2009	0.82	0.82	0.82
2010	0.71	0.80	0.60
2011	0.63	0.70	0.55

Figure 4: Distributions of team ratings from the 2011 FE.P.1 model. The dashed line corresponds to FCS teams, the solid line to FBS teams.



4.5 ML vs. REML

For now, we will only consider the sensitivity to the choice of ML versus REML estimation when using PQL in SAS. Under the model PQL.P.1, none of the top 16 teams from 2008-2011 differ between the ML and the REML rankings (not shown). This is not surprising since our model includes only one fixed effect, the FCS effect, and around 240 different levels of the random effect, corresponding to the FBS and FCS teams. Unlike a one-way random effects model, it is not clear what the expected downward bias in the ML estimates of σ_t^2 should be in this multi-membership random effects setting. However, we found the estimates from two methods to be nearly identical. We include estimates from the two methods in Table 3 so that the difference between the ML and REML estimates may be compared to the differences resulting from different integral approximations. Of course, the difference between ML and REML estimates would grow if additional fixed effects were added to the model.

5 Conclusion

We have proposed a generalization of the model developed by Mease (2003) and tested our model's rankings for sensitivity to several modeling choices. Ideally, this type of sensitivity analysis should be performed whenever a generalized

Table 3: Estimates for σ_t^2 from PQL.P.1

Year	ML	REML
2008	0.4763	0.4768
2009	0.5077	0.5087
2010	0.4405	0.4415
2011	0.4206	0.4216

linear mixed model is used. The sensitivity will likely depend on the random effects structure of different models. The downward bias in PQL estimates of variance components has been well documented (Breslow and Lin, 1995), but not as much attention has been given to differences in EBLUP orderings resulting from using different orders of integral approximation.

Harville (2003) discusses seven criteria for an appropriate ranking-model: accuracy, appropriateness, impartiality, unobtrusiveness, nondisruptiveness, verifiability, and comprehensibility. For practical purposes, the choice between Mease’s model and our model FE.P.2 represents a trade-off between comprehensibility and accuracy. The computational effort required to obtain the rankings from FE.P.2 is much greater than that required for Mease’s model. Furthermore, the lack of a closed-form objective function obfuscates the relationship between the data and the rankings. However, by modeling a separate FCS population and using a fully exponential Laplace approximation, FE.P.2 makes use of additional data and provides the capacity to accurately estimate the FBS population variance, whereas Mease relies on a fixed population variance.

The sensitivity of the EBLUPs to different methods of approximating the marginal likelihood may be of interest in other settings, including in the use of value-added models for teacher assessment. Value-added models evaluate teachers based on the performance of their students. When the measures of student performance are categorical (Broatch and Lohr, 2012), the analysis of the sensitivity to the choice of approximation of the marginal likelihood that we discuss in this paper may be relevant.

Without changing the mean or random effects structures, our rankings shifted with different choices of modeling assumptions. The resulting changes in team ratings are small relative to the standard errors of the ratings, but could have implications for which teams are assigned to which bowls. Bowls are assigned based on the point estimates of the team rankings: an undefeated,

third ranked team would probably take little consolation in being told that their rating is not significantly lower than those of the top two teams.

The large confidence intervals associated with the team ratings suggest that the sensitivity of the rankings to the modeling assumptions is due at least in part to the limited information available to the model: around 12 binary outcomes on 240 or so subjects. Given this limited information, it seems unreasonable to expect these models to be capable of identifying the two best teams in a division. The limited accuracy of the models due to their restriction to the use of binary game outcomes is one of the reasons that lead Stern (2006) to call for a boycott of the BCS by quantitative analysts. Models for binary game outcomes may provide a rough guide for the classification of teams, but the fickle nature of their rankings should be kept in mind.

A Tables and Graphs

Note: For the plots of the rankings (e.g. Figure 5), the best teams appear in the bottom left corner of each graph, corresponding to rankings near 1. By contrast, the best teams in the plots of the ratings appear in the top right corner, corresponding to larger ratings (e.g. Figure 6).

Table 4: 2008 Division I-A Pre-Bowl Rankings and Ratings, sorted by FE.P.0

BCS	Team		PQL.P.0		LA.P.0		FE.P.0		FE.L.0
1	Oklahoma	1	1.347	1	1.369	1	1.714	1	2.911
6	Utah	2	1.269	2	1.289	2	1.631	2	2.770
3	Texas	3	1.248	3	1.267	3	1.582	3	2.670
7	Texas Tech	6	1.156	5	1.176	4	1.494	4	2.530
2	Florida	4	1.196	4	1.212	5	1.465	5	2.524
9	Boise St.	5	1.147	6	1.162	6	1.438	6	2.434
4	Alabama	7	1.104	7	1.119	7	1.369	7	2.301
5	Southern Cal	8	1.043	8	1.057	8	1.284	8	2.198
8	Penn St.	9	1.007	9	1.020	9	1.237	9	2.136
10	Ohio St.	10	0.890	10	0.902	10	1.091	10	1.846
11	TCU	12	0.834	12	0.847	11	1.059	11	1.785
12	Cincinnati	11	0.840	11	0.850	12	1.016	12	1.753
22	Ball St.	13	0.808	13	0.815	13	0.953	13	1.648
13	Oklahoma St.	16	0.686	15	0.698	14	0.877	14	1.481
15	Georgia	14	0.711	14	0.722	15	0.875	15	1.480
16	BYU	15	0.684	16	0.692	16	0.850	16	1.425
18	Michigan St.	17	0.662	17	0.670	17	0.799	19	1.349
14	Georgia Tech	19	0.633	19	0.644	18	0.796	17	1.383
20	Pittsburgh	18	0.650	18	0.657	19	0.760	18	1.350
17	Oregon	20	0.589	20	0.597	20	0.720	22	1.208
σ_t^2			0.52		0.54		0.76		2.19

Figure 5: 2008 Division I-A Pre-Bowl Rankings

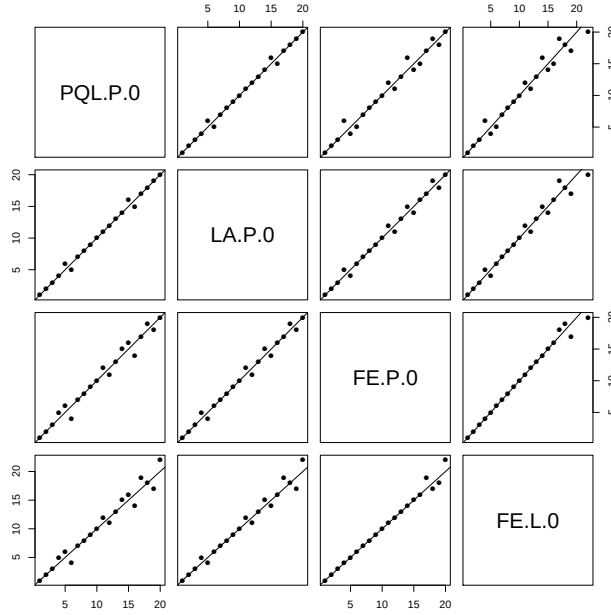


Figure 6: 2008 Division I-A Pre-Bowl Ratings. Lines have slope 1 and intercept 0.

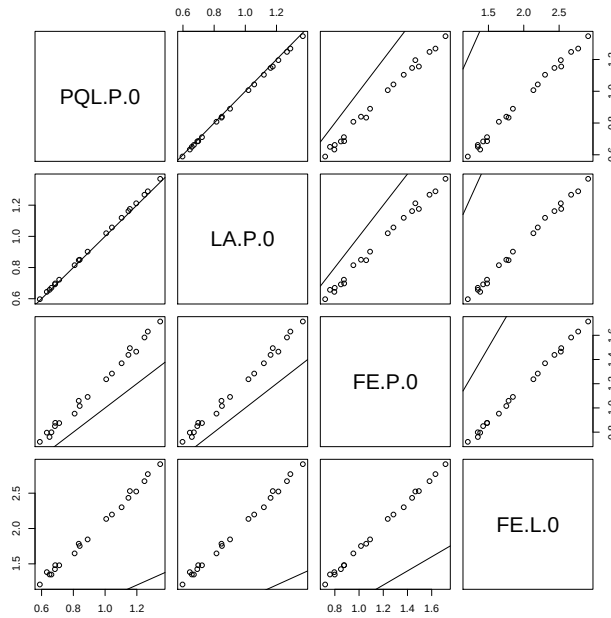


Table 5: 2008 Division I-A Pre-Bowl Rankings and Ratings, sorted by FE.P.2

BCS	Team		FE.P.2		FE.P.1		FE.P.0		Mease
1	Oklahoma	1	1.589	1	1.700	1	1.714	1	1.648
6	Utah	2	1.513	2	1.619	2	1.631	2	1.538
3	Texas	3	1.469	3	1.569	3	1.582	3	1.512
2	Florida	4	1.381	5	1.459	5	1.465	5	1.399
7	Texas Tech	5	1.375	4	1.480	4	1.494	4	1.433
9	Boise St.	6	1.342	6	1.422	6	1.438	6	1.338
4	Alabama	7	1.290	7	1.367	7	1.369	7	1.296
5	Southern Cal	8	1.207	8	1.275	8	1.284	8	1.214
8	Penn St.	9	1.162	9	1.228	9	1.237	9	1.169
10	Ohio St.	10	1.024	10	1.082	10	1.091	10	1.033
11	TCU	11	0.981	11	1.050	11	1.059	11	1.003
12	Cincinnati	12	0.956	12	1.007	12	1.016	12	0.965
22	Ball St.	13	0.912	13	0.947	13	0.953	13	0.880
15	Georgia	14	0.820	14	0.871	15	0.875	15	0.835
13	Oklahoma St.	15	0.804	15	0.866	14	0.877	14	0.837
16	BYU	16	0.800	16	0.848	16	0.850	16	0.794
18	Michigan St.	17	0.754	18	0.794	17	0.799	18	0.755
14	Georgia Tech	18	0.744	17	0.799	18	0.796	17	0.766
20	Pittsburgh	19	0.722	19	0.754	19	0.760	19	0.724
19	Virginia Tech	20	0.678	20	0.726	21	0.719	21	0.692

Karl: Sensitivity of College Football Rankings

Figure 7: 2008 Division I-A Pre-Bowl Rankings

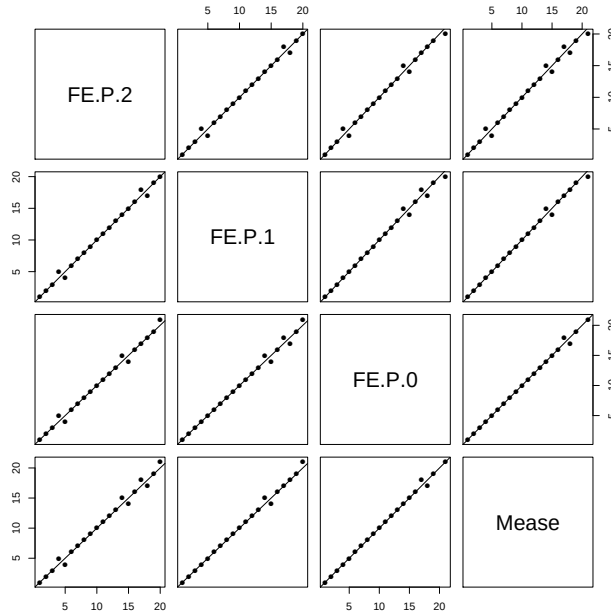


Figure 8: 2008 Division I-A Pre-Bowl Ratings

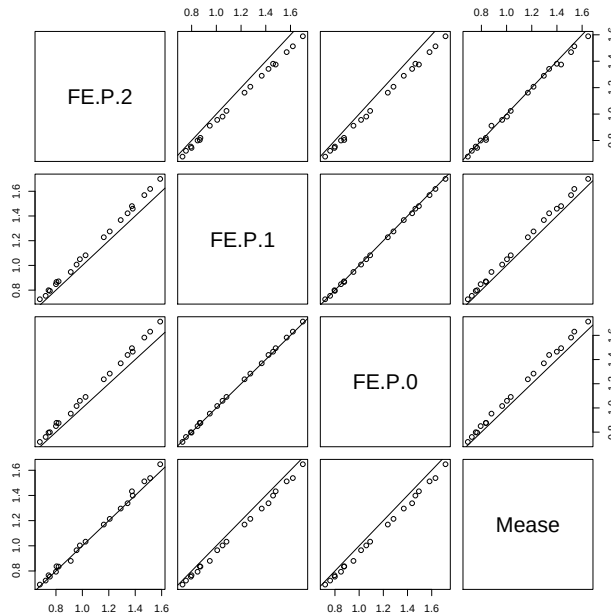


Table 6: 2009 Division I-A Pre-Bowl Rankings and Ratings, sorted by FE.P.0

BCS	Team		PQL.P.0		LA.P.0		FE.P.0		FE.L.0
1	Alabama	1	1.592	1	1.618	1	2.064	1	3.543
3	Cincinnati	3	1.390	3	1.412	2	1.789	2	3.077
2	Texas	2	1.413	2	1.431	3	1.781	3	3.043
5	Florida	4	1.309	4	1.330	4	1.675	4	2.855
4	TCU	5	1.262	5	1.280	5	1.624	5	2.799
6	Boise St.	6	1.256	6	1.273	6	1.616	6	2.777
7	Oregon	7	1.041	7	1.056	7	1.303	7	2.247
9	Georgia Tech	8	0.904	8	0.916	8	1.112	8	1.948
10	Iowa	9	0.864	9	0.875	9	1.051	9	1.835
8	Ohio St.	10	0.832	10	0.841	10	1.000	10	1.796
12	LSU	12	0.779	12	0.791	11	0.997	11	1.718
11	Virginia Tech	11	0.786	11	0.797	12	0.970	12	1.686
13	Penn St.	13	0.760	13	0.768	13	0.921	14	1.580
14	BYU	14	0.754	14	0.763	14	0.920	13	1.580
16	West Virginia	15	0.684	15	0.694	15	0.844	15	1.477
15	Miami (FL)	16	0.681	16	0.691	16	0.840	16	1.466
19	Oklahoma St.	17	0.654	17	0.662	17	0.798	18	1.379
17	Pittsburgh	18	0.637	18	0.645	18	0.776	17	1.392
20	Arizona	21	0.581	21	0.591	19	0.746	19	1.306
18	Oregon St.	23	0.563	22	0.574	20	0.733	20	1.261
σ_t^2			0.57		0.59		0.85		2.54

Karl: Sensitivity of College Football Rankings

Figure 9: 2009 Division I-A Pre-Bowl Rankings

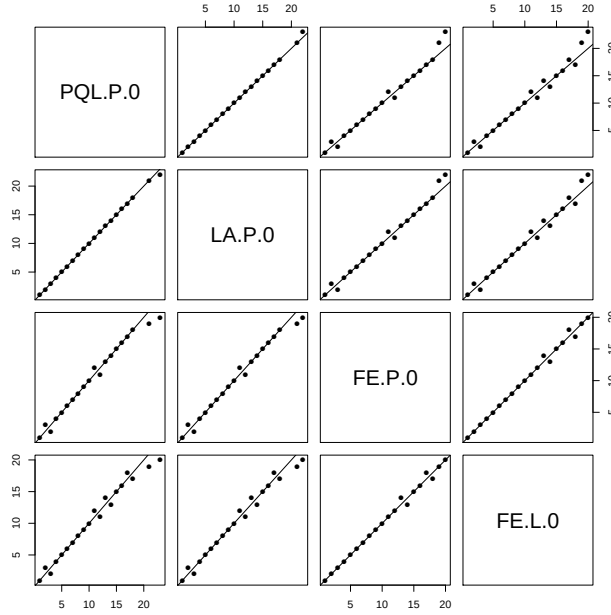


Figure 10: 2009 Division I-A Pre-Bowl Ratings

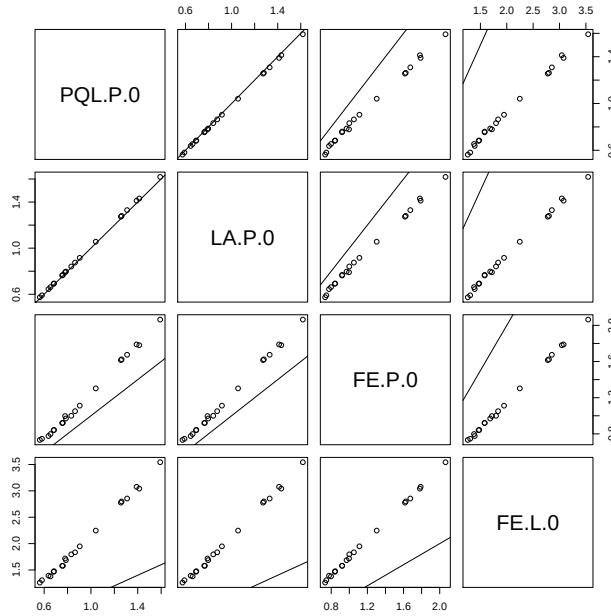


Table 7: 2009 Division I-A Pre-Bowl Rankings and Ratings, sorted by FE.P.2

BCS	Team		FE.P.2		FE.P.1		FE.P.0		Mease
1	Alabama	1	2.019	1	2.016	1	2.064	1	1.883
3	Cincinnati	2	1.745	2	1.742	2	1.789	2	1.616
2	Texas	3	1.742	3	1.740	3	1.781	3	1.609
5	Florida	4	1.635	4	1.633	4	1.675	4	1.527
4	TCU	5	1.594	5	1.592	5	1.624	5	1.450
6	Boise St.	6	1.576	6	1.574	6	1.616	6	1.440
7	Oregon	7	1.271	7	1.270	7	1.303	7	1.191
9	Georgia Tech	8	1.103	8	1.102	8	1.112	8	1.019
10	Iowa	9	1.021	9	1.021	9	1.051	9	0.959
8	Ohio St.	10	0.977	10	0.976	10	1.000	10	0.914
11	Virginia Tech	11	0.965	11	0.965	12	0.970	12	0.890
12	LSU	12	0.961	12	0.960	11	0.997	11	0.908
13	Penn St.	13	0.907	13	0.907	13	0.921	13	0.834
14	BYU	14	0.896	14	0.896	14	0.920	14	0.833
15	Miami (FL)	15	0.835	15	0.834	16	0.840	16	0.769
16	West Virginia	16	0.816	16	0.816	15	0.844	15	0.775
19	Oklahoma St.	17	0.768	17	0.768	17	0.798	17	0.732
17	Pittsburgh	18	0.746	18	0.745	18	0.776	18	0.714
20	Arizona	19	0.717	19	0.717	19	0.746	19	0.678
18	Oregon St.	20	0.702	20	0.701	20	0.733	20	0.666

Figure 11: 2009 Division I-A Pre-Bowl Rankings

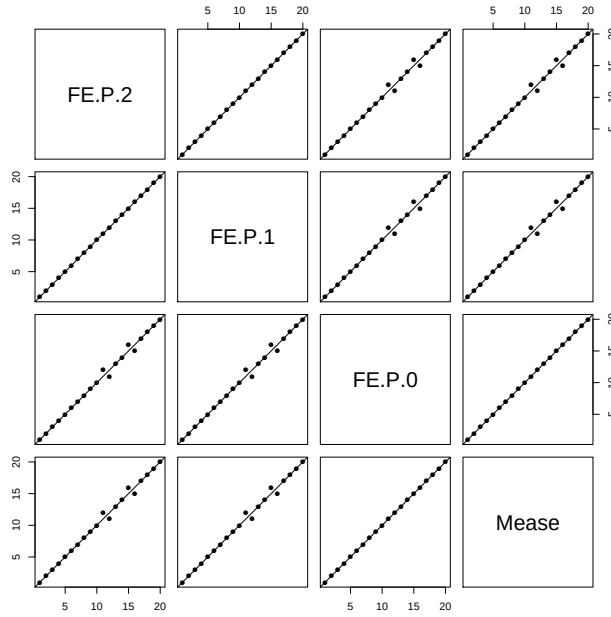


Figure 12: 2009 Division I-A Pre-Bowl Ratings

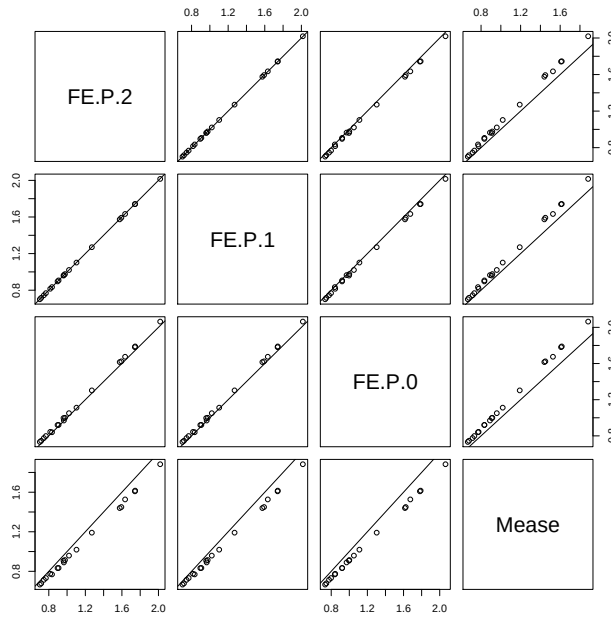


Table 8: 2010 Division I-A Pre-Bowl Rankings and Ratings, sorted by FE.P.0

BCS	Team		PQL.P.0		LA.P.0		FE.P.0		FE.L.0
1	Auburn	1	1.546	1	1.573	1	1.996	1	3.423
2	Oregon	2	1.326	2	1.348	2	1.725	2	2.941
3	TCU	3	1.287	3	1.307	3	1.659	3	2.838
4	Stanford	5	1.099	5	1.116	4	1.400	4	2.370
7	Oklahoma	4	1.111	4	1.128	5	1.381	5	2.345
5	Wisconsin	8	1.073	7	1.089	6	1.364	6	2.316
6	Ohio St.	6	1.085	6	1.099	7	1.347	9	2.277
8	Arkansas	10	1.036	10	1.054	8	1.331	8	2.288
9	Michigan St.	7	1.073	8	1.088	9	1.324	7	2.299
10	Boise St.	9	1.052	9	1.066	10	1.303	10	2.269
11	LSU	11	1.005	11	1.022	11	1.280	11	2.185
15	Nevada	12	0.977	12	0.989	12	1.214	12	2.094
12	Missouri	13	0.973	13	0.988	13	1.210	13	2.092
14	Oklahoma St.	14	0.958	14	0.972	14	1.197	14	2.028
17	Texas A&M	15	0.858	15	0.873	15	1.095	15	1.857
16	Alabama	16	0.795	16	0.809	16	1.030	16	1.761
19	Utah	17	0.793	17	0.804	17	0.983	18	1.681
18	Nebraska	18	0.781	18	0.794	18	0.982	17	1.701
20	South Carolina	19	0.665	19	0.675	19	0.833	20	1.471
13	Virginia Tech	20	0.658	20	0.663	20	0.749	19	1.631
σ_t^2			0.54		0.56		0.81		2.37

Karl: Sensitivity of College Football Rankings

Figure 13: 2010 Division I-A Pre-Bowl Rankings

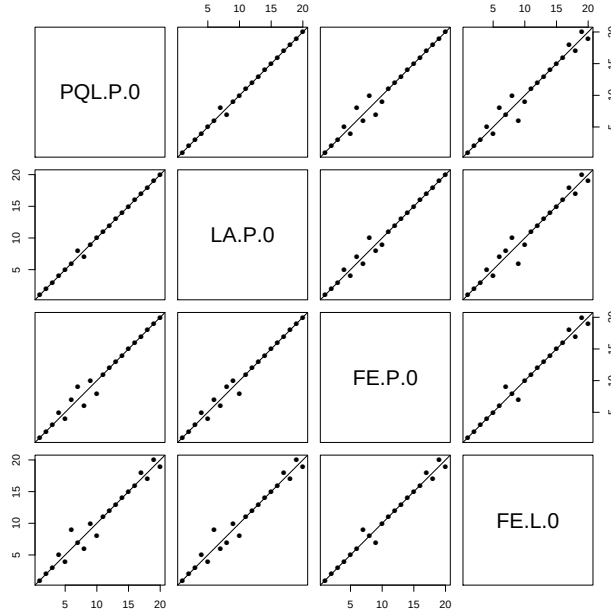


Figure 14: 2010 Division I-A Pre-Bowl Ratings

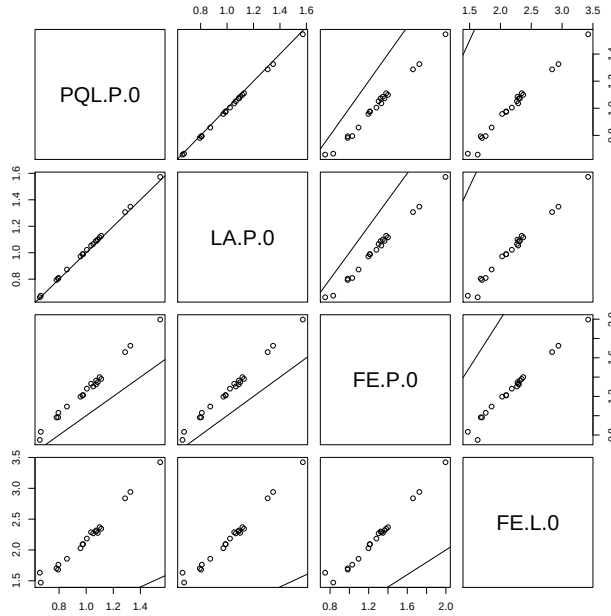


Table 9: 2010 Division I-A Pre-Bowl Rankings and Ratings, sorted by FE.P.2

BCS	Team		FE.P.2		FE.P.1		FE.P.0		Mease
1	Auburn	1	1.983	1	1.871	1	1.996	1	1.867
2	Oregon	2	1.711	2	1.612	2	1.725	2	1.596
3	TCU	3	1.644	3	1.556	3	1.659	3	1.524
4	Stanford	4	1.387	4	1.313	4	1.400	4	1.303
7	Oklahoma	5	1.371	5	1.307	5	1.381	5	1.292
5	Wisconsin	6	1.347	6	1.278	6	1.364	6	1.259
6	Ohio St.	7	1.335	7	1.276	7	1.347	7	1.245
8	Arkansas	8	1.320	9	1.245	8	1.331	8	1.243
9	Michigan St.	9	1.307	8	1.251	9	1.324	9	1.229
10	Boise St.	10	1.296	10	1.241	10	1.303	10	1.200
11	LSU	11	1.272	11	1.205	11	1.280	11	1.193
15	Nevada	12	1.210	12	1.159	12	1.214	14	1.113
12	Missouri	13	1.202	13	1.145	13	1.210	12	1.132
14	Oklahoma St.	14	1.190	14	1.133	14	1.197	13	1.114
17	Texas A&M	15	1.091	15	1.032	15	1.095	15	1.023
16	Alabama	16	1.021	16	0.961	16	1.030	16	0.962
18	Nebraska	17	0.974	18	0.925	18	0.982	17	0.918
19	Utah	18	0.971	17	0.928	17	0.983	18	0.908
20	South Carolina	19	0.819	19	0.778	19	0.833	19	0.783
13	Virginia Tech	20	0.783	20	0.768	20	0.749	20	0.704

Figure 15: 2010 Division I-A Pre-Bowl Rankings

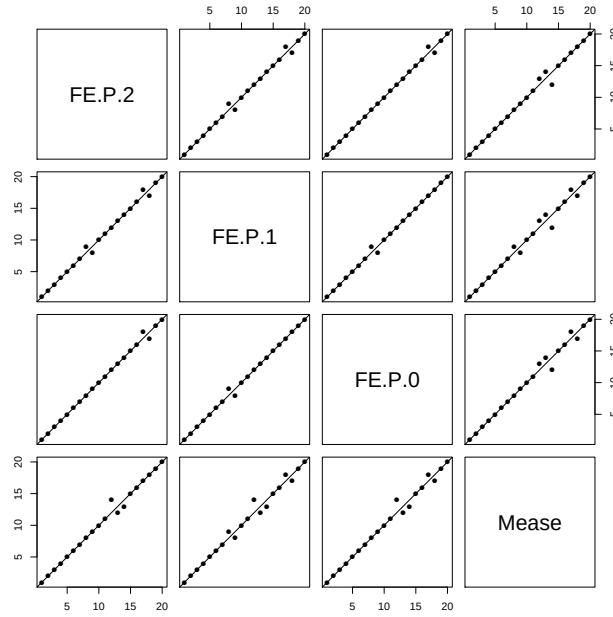


Figure 16: 2010 Division I-A Pre-Bowl Ratings

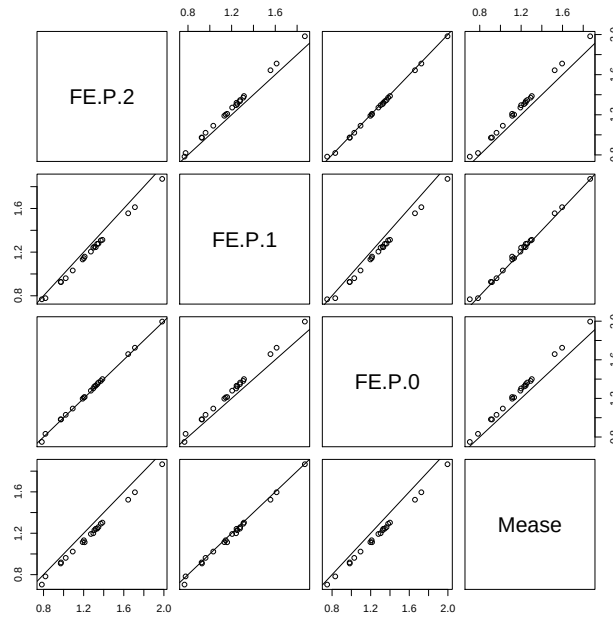


Table 10: 2011 Division I-A Pre-Bowl Rankings and Ratings, sorted by FE.P.0

BCS	Team		PQL.P.0		LA.P.0		FE.P.0		FE.L.0
1	LSU	1	1.626	1	1.651	1	2.076	1	3.544
2	Alabama	3	1.233	3	1.252	2	1.572	3	2.667
3	Oklahoma St.	2	1.267	2	1.286	3	1.565	2	2.749
4	Stanford	4	1.093	4	1.107	4	1.347	4	2.295
7	Boise St.	5	1.069	5	1.082	5	1.308	5	2.231
6	Arkansas	8	0.993	7	1.010	6	1.276	6	2.168
8	Kansas St.	6	1.010	6	1.026	7	1.268	7	2.157
5	Oregon	7	0.996	8	1.010	8	1.233	8	2.086
19	Houston	9	0.975	9	0.985	9	1.166	9	2.000
9	South Carolina	10	0.930	10	0.943	10	1.151	10	1.962
11	Virginia Tech	11	0.921	11	0.931	11	1.103	13	1.870
14	Oklahoma	14	0.878	13	0.892	12	1.092	11	1.903
probation	Southern Cal	12	0.898	12	0.909	13	1.083	12	1.892
12	Baylor	16	0.846	16	0.860	14	1.069	14	1.840
13	Michigan	13	0.879	14	0.888	15	1.042	16	1.799
16	Georgia	17	0.834	17	0.847	16	1.042	17	1.762
10	Wisconsin	15	0.857	15	0.867	17	1.031	15	1.819
18	TCU	18	0.764	18	0.773	18	0.925	18	1.596
15	Clemson	19	0.757	19	0.767	19	0.916	19	1.590
17	Michigan St.	20	0.708	20	0.717	20	0.863	21	1.480
σ_t^2			0.55		0.57		0.80		2.33

Figure 17: 2011 Division I-A Pre-Bowl Rankings

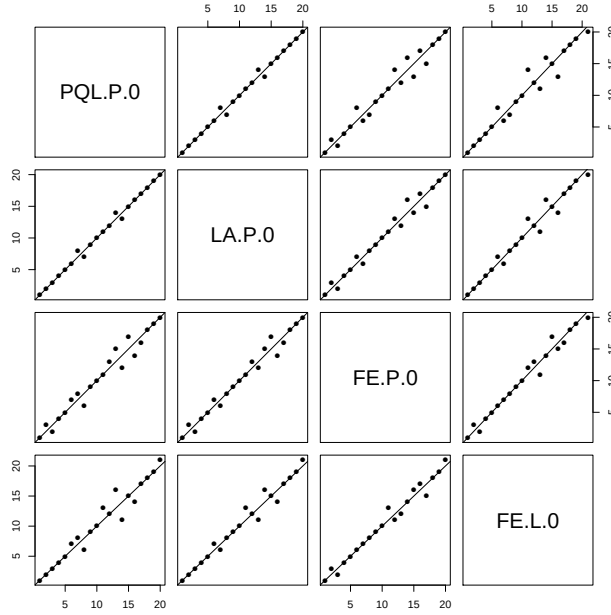


Figure 18: 2011 Division I-A Pre-Bowl Ratings

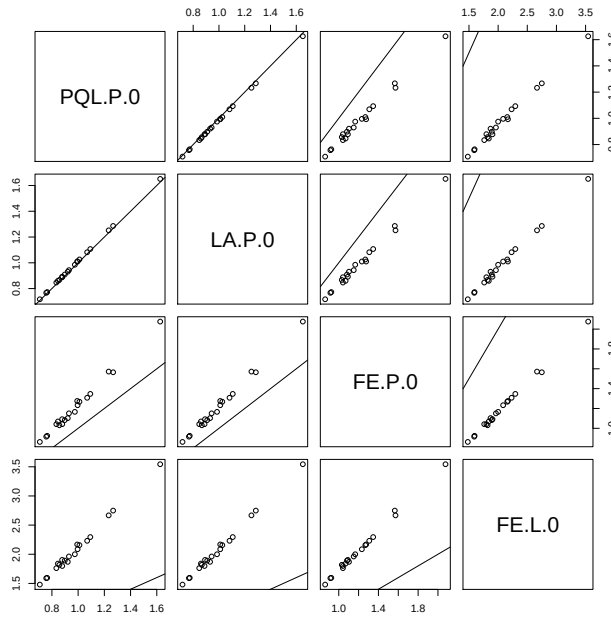


Table 11: 2011 Division I-A Pre-Bowl Rankings and Ratings, sorted by FE.P.2

BCS	Team		FE.P.2		FE.P.1		FE.P.0		Mease
1	LSU	1	1.944	1	1.463	1	2.076	1	1.954
3	Oklahoma St.	2	1.482	2	1.152	3	1.565	2	1.477
2	Alabama	3	1.474	3	1.108	2	1.572	3	1.470
4	Stanford	4	1.263	4	0.986	4	1.347	4	1.254
7	Boise St.	5	1.243	5	0.980	5	1.308	5	1.209
8	Kansas St.	6	1.192	6	0.906	7	1.268	7	1.194
6	Arkansas	7	1.185	9	0.879	6	1.276	6	1.196
5	Oregon	8	1.151	8	0.891	8	1.233	8	1.154
19	Houston	9	1.106	7	0.897	9	1.166	10	1.077
9	South Carolina	10	1.077	11	0.831	10	1.151	9	1.080
11	Virginia Tech	11	1.044	10	0.837	11	1.103	12	1.027
14	Oklahoma	12	1.029	14	0.789	12	1.092	11	1.032
probation	Southern Cal	13	1.023	12	0.815	13	1.083	13	1.016
12	Baylor	14	1.012	16	0.751	14	1.069	14	1.008
13	Michigan	15	0.994	13	0.805	15	1.042	16	0.973
10	Wisconsin	16	0.979	15	0.780	17	1.031	17	0.961
16	Georgia	17	0.971	17	0.740	16	1.042	15	0.978
18	TCU	18	0.877	18	0.695	18	0.925	19	0.853
15	Clemson	19	0.862	19	0.679	19	0.916	18	0.859
17	Michigan St.	20	0.817	20	0.638	20	0.863	20	0.802

Figure 19: 2011 Division I-A Pre-Bowl Rankings

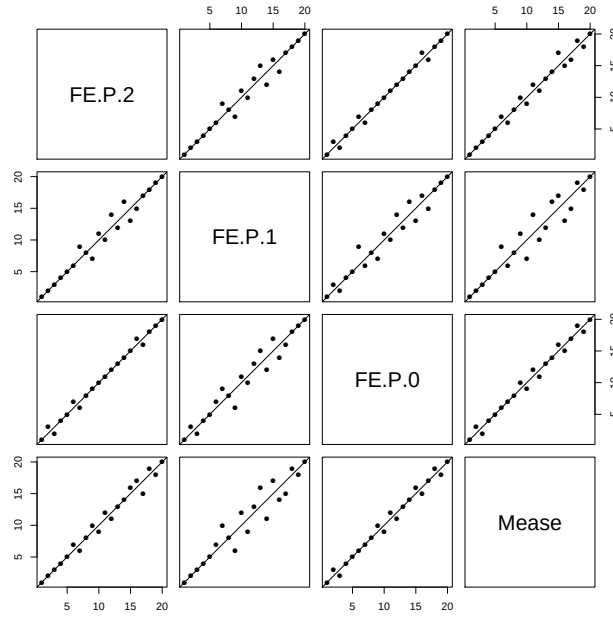
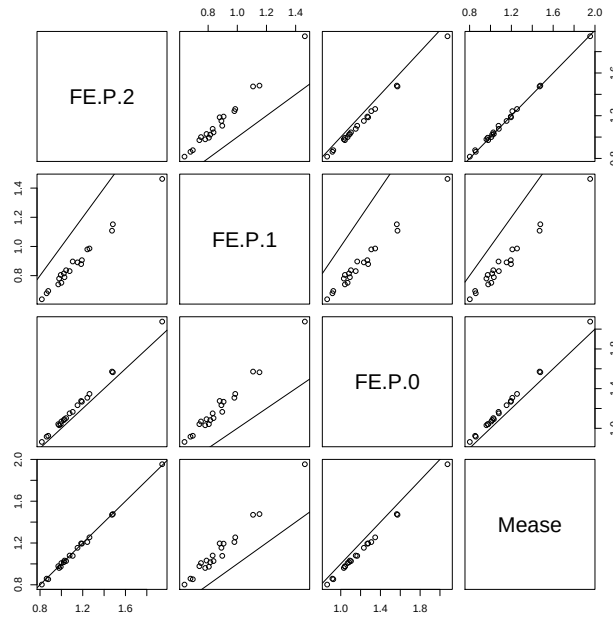


Figure 20: 2011 Division I-A Pre-Bowl Ratings



B SAS Code

The SAS code to produce the 2008 PQL.P.1 rankings appears in Listing 1. The EFFECT statement allows us to specify our multi-membership design matrix by assigning a coefficient of 1, stored in the “H” column, to the home team in each game, and a coefficient of -1, stored in the “A” column, to the visiting team. The EFFECT statement recognizes that each team appears in both the “home” and “away” variables, and assigns only one column to each team in the constructed effect we have called “matchup.” See Table 12 for the data used by this SAS procedure.

Listing 1: Example SAS Code

```
PROC GLIMMIX data=d2008 method=mspl;
    class home away home_win;
    effect matchup = MULTIMEMBER(home away/weight=(H A));
    model home_win (descending)=fcs/noint dist=binary
        link=probit solution;
    random matchup /solution;
    ods output solutionr=ratings2008;
run;
```

Table 12: SAS Data Set d2008

home	Game Date	away	home_score	away_score	fcs	H	A	home_win
Ball St.	8/28/2008	Northeastern	48	14	1	1	-1	1
Baylor	8/28/2008	Wake Forest	13	41	0	1	-1	0
Buffalo	8/28/2008	UTEP	42	17	0	1	-1	1
:	:	:	:	:	:	:	:	:

References

- Allison, P. D. (2008), “Convergence Failures in Logistic Regression,” SAS Global Forum 2008, no. 360-2008, www2.sas.com/proceedings/forum2008/360-2008.pdf.
- Annis, D. H. and Craig, B. A. (2005), “Hybrid Paired Comparison Analysis, with Applications to the Ranking of College Football Teams,” *Journal of Quantitative Analysis in Sports*, 1, Iss. 1, Article 3.
- Bowl Championship Series (2011), “BCS selection procedures,” Press Release, <http://www.bcsfootball.org/news/story?id=4819597>.

- Bradley, R. A. and Terry, M. E. (1952), "The Rank Analysis of Incomplete Block Diagrams. I. The Method of Paired Comparisons," *Biometrika*, 39, 324–345.
- Breslow, N. E. and Clayton, D. G. (1993), "Approximate Inference in Generalized Linear Mixed Models," *Journal of the American Statistical Association*, 88, 9–25.
- Breslow, N. E. and Lin, X. (1995), "Bias Correction in Generalised Linear Mixed Models with a Single Component of Dispersion," *Biometrika*, 82, 81–91.
- Broatch, J. and Lohr, S. (2012), "Multidimensional Assessment of Value Added by Teachers to Real-World Outcomes," *Journal of Educational and Behavioral Statistics*, 37, 256–277.
- Browne, W. J., Goldstein, H., and Rasbash, J. (2001), "Multiple Membership Multiple Classification (MMMC) Models," *Statistical Modelling*, 1, 103–124.
- Demidenko, E. (2004), *Mixed Models Theory and Applications*, Hoboken: Wiley-Interscience.
- Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977), "Maximum Likelihood from Incomplete Data Via the EM Algorithm," *Journal of the Royal Statistical Society, Series B*, 39, 1–38.
- Draper, D. (1995), "Inference and Hierarchical Modeling in the Social Sciences," *Journal of Educational and Behavioral Statistics*, 20, 115–117.
- Eichelberger, C. (2012), "LSU Football Coach Les Miles Has \$5.7 Million Riding on BCS National Title," *Bloomberg*, <http://www.bloomberg.com/news/2012-01-09/lsu-football-coach-les-miles-has-5-7-million-riding-on-bcs-national-title.html>.
- Evans, M. and Swartz, T. (1995), "Methods for Approximating Integrals in Statistics with Special Emphasis on Bayesian Integration Problems," *Statistical Science*, 10, 254–272.
- Gill, R. and Keating, J. (2009), "Assessing Methods for College Football Rankings," *Journal of Quantitative Analysis in Sports*, 5, Iss. 2, Article 3.
- Harville, D. (1977a), "The Use of Linear-Model Methodology to Rate High School or College Football Teams," *Journal of the American Statistical Association*, 72, 278–289.

- Harville, D. A. (1977b), “Maximum Likelihood Approaches to Variance Component Estimation and to Related Problems,” *Journal of the American Statistical Association*, 72, 320–338.
- (2003), “The Selection or Seeding of College Basketball or Football Teams for Postseason Competition,” *Journal of the American Statistical Association*, 98, 17–27.
- Karl, A. T. (2012a), “A Correlated Random-Effects Model for Nonignorable Missing Data in Value-Added Assessment of Teacher Effects,” Ph.D. thesis, Arizona State University.
- (2012b), “Personal Website,” <https://sites.google.com/site/andrewtkarl>.
- Karl, A. T., Yang, Y., and Lohr, S. L. (2012), “Efficient Maximum Likelihood Estimation of Multiple Membership Linear Mixed Models, with an Application to Educational Value-Added Assessments,” *Submitted*.
- Keener, J. P. (1993), “The Perron-Frobenius Theorem and the Ranking of Football Teams,” *SIAM Review*, 35, 80–93.
- Laird, N. M. and Ware, J. H. (1982), “Random-Effects Models for Longitudinal Data,” *Biometrics*, 38, 963–974.
- Littell, R. C., Milliken, G. A., Stroup, W. W., Wolfinger, R. D., and Schabenberger, O. (2006), *SAS for Mixed Models*, Cary, NC: SAS Institute, Inc., 2nd ed.
- McCulloch, C. E. (1994), “Maximum Likelihood Variance Components Estimation for Binary Data,” *Journal of the American Statistical Association*, 89, 330–335.
- McLachlan, G. J. and Krishnan, T. (2008), *The EM Algorithm and Extensions*, Hoboken: John Wiley & Sons, 2nd ed.
- Mease, D. (2003), “A Penalized Maximum Likelihood Approach for the Ranking of College Football Teams Independent of Victory Margins,” *The American Statistician*, 57, 241–248.
- (2012), “Personal Website,” <http://www.davemease.com/football/>.

- Mosteller, F. (1951), "Remarks on the Method of Paired Comparisons: I. The Least Squares Solution Assuming Equal Standard Deviations and Equal Correlations," *Psychometrika*, 16, 3–9.
- NCAA (2012), "NCAA Football Statistics," <http://www.ncaa.org/wps/wcm/connect/public/NCAA/Resources/Stats/Football/index.html>.
- R Development Core Team (2012), *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, Austria, ISBN 3-900051-07-0.
- Rizopoulos, D., Verbeke, G., and Lesaffre, E. (2009), "Fully Exponential Laplace Approximations for the Joint Modelling of Survival and Longitudinal Data," *Journal of the Royal Statistical Society, Series B*, 71, 637–654.
- Shun, Z. and McCullagh, P. (1995), "Laplace Approximation of High Dimensional Integrals," *Journal of the Royal Statistical Society, Series B*, 57, 749–760.
- Steele, B. M. (1996), "A Modified EM Algorithm for Estimation in Generalized Mixed Models," *Biometrics*, 52, 1295–1310.
- Stern, H. S. (2006), "In Favor of A Quantitative Boycott of the Bowl Championship Series," *Journal of Quantitative Analysis in Sports*, 2, Iss. 1, Article 4.
- Stern, H. S., Massey, K., Harville, D. A., Billingsley, R., Colley, W., and Mease, D. (2004), "Statistics and the College Football Championship," *The American Statistician*, 58, 179–195.
- Thurstone, L. L. (1927), "A Law of Comparative Judgment," *Psychological Review*, 34, 273–286.
- Tierney, L., Kass, R. E., and Kadane, J. B. (1989), "Fully Exponential Laplace Approximations to Expectations and Variances of Nonpositive Functions," *Journal of the American Statistical Association*, 84, 710–716.
- Wang, W., Johnston, R., and Jones, K. (2011), "Home Advantage in American College Football Games: A Multilevel Modelling Approach," *Journal of Quantitative Analysis in Sports*, 7, Iss. 3, Article 23.
- West, B. T. (2006), "A Simple and Flexible Rating Method for Predicting Success in the NCAA Basketball Tournament," *Journal of Quantitative Analysis in Sports*, 2, Iss. 3, Article 3.

- West, B. T. and Lamsal, M. (2008), “A New Application of Linear Modeling in the Prediction of College Football Bowl Outcomes and the Development of Team Ratings,” *Journal of Quantitative Analysis in Sports*, 4, Iss. 3, Article 3.
- Wolfinger, R. and O’Connell, M. (1993), “Generalized Linear Mixed Models: A Pseudo-likelihood Approach,” *Journal of Statistical Computation and Simulation*, 48, 233–243.